

Problem Statement: Agentic AI LinkedIn Profile Research Automation Using n8n

Background

Organizations frequently need to identify and collect LinkedIn company profiles based on website domains for sales prospecting, recruitment, market research, competitor analysis, and business intelligence. Performing this task manually is time-consuming, repetitive, and prone to human error, especially when dealing with hundreds or thousands of domains.

Problem

Given a list of company domains, there is no automated mechanism to accurately search, identify, validate, and store the corresponding LinkedIn company profiles in a structured format. Manual searching across search engines and LinkedIn consumes significant effort and reduces operational efficiency.

Objective

Design and implement an **Agentic AI workflow using n8n** that automatically retrieves LinkedIn company profile information from company domains and stores the results in Google Sheets for further analysis and reporting.

Functional Requirements

1. Accept a company domain as input.
2. Perform an intelligent web search using the Serper API.
3. Identify the most relevant LinkedIn company profile associated with the domain.
4. Extract relevant information such as:
 - Company Name
 - LinkedIn Company URL
 - Job Title (if available)
 - Search Status
5. Validate that the extracted URL belongs to LinkedIn.
6. Store the extracted information in Google Sheets.
7. Process multiple domains automatically through a loop mechanism.
8. Handle missing or invalid results gracefully.
9. Update the status of each search operation (Found / Not Found / Error).

Proposed Solution

The solution utilizes an **Agentic AI architecture in n8n** consisting of:

Manual Trigger



HTTP Request (Serper API)



Code Node (Data Extraction & Validation)



Loop Over Items



Google Sheets

The workflow autonomously searches for LinkedIn profiles, validates results, extracts relevant information, and updates a centralized Google Sheet without manual intervention.

Expected Output

Domain	LinkedIn URL	Name	Job Title	Status
openai.com	linkedin.com/company/openai	OpenAI	N/A	Found
microsoft.com	linkedin.com/company/microsoft	Microsoft	N/A	Found

Benefits

- Eliminates manual LinkedIn searching.
- Improves data collection efficiency.
- Reduces human errors.
- Enables scalable company research.
- Demonstrates Agentic AI automation using n8n, APIs, and cloud-based data storage.

Technologies Used

- n8n Workflow Automation
- Serper API
- JavaScript Code Nodes
- Google Sheets API
- Agentic AI Workflow Design